Smart Study Recommender Using Machine Learning

Course: Fundamentals of AI and ML

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1. Introduction

Artificial Intelligence (AI) and Machine Learning (ML) are widely used to analyze data and make intelligent predictions.
In education, these technologies can help students identify weak subjects and improve their study strategies.

This project builds a simple machine learning system that analyzes a student's scores in Physics, Chemistry, and Mathematics
and recommends which subject they should focus on. The goal is to demonstrate how AI concepts learned in the course can be
applied to a practical real-world problem.

2. Problem Statement

Students often struggle to identify which subject they should prioritize while studying. Without proper analysis of their performance, they may spend time revising subjects they are already good at.

The objective of this project is to develop a machine learning model that analyzes student scores and recommends
the subject that requires the most attention.

3. Objectives

• Apply machine learning concepts learned in the course.
• Build a simple predictive model using student score data.
• Automatically determine the weakest subject for a student.

• Provide recommendations that help students focus their study efforts.
• Demonstrate the workflow of a machine learning project.

4. Dataset

The dataset used in this project contains student scores for three subjects:
Physics, Chemistry, and Mathematics.

Each row represents a student's performance. The system analyzes these scores and determines the subject with
the lowest score, which becomes the recommended subject to focus on.

Example dataset format:

Physics	Chemistry	Maths
45	78	52
82	60	88
55	50	48

The dataset was expanded to include many records to improve the learning ability of the machine learning model.

5. Methodology

The project follows a typical machine learning workflow:

1. Data Collection – A dataset of student scores was created.
2. Data Preprocessing – The dataset was cleaned and prepared for model training.
3. Label Generation – The system automatically identifies the weakest subject.
4. Model Training – A Decision Tree classifier is trained on the dataset.
5. Model Evaluation – Accuracy is calculated using test data.
6. Prediction – The model predicts the subject needing the most improvement.

6. Tools and Technologies

The following tools and technologies were used in the project:

• Python
• Pandas – Data handling
• Scikit-learn – Machine learning models
• Joblib – Saving trained models
• Matplotlib – Data visualization
• Streamlit – Web interface for predictions
• GitHub – Version control and project hosting

7. Implementation

The implementation consists of several components:

1. Training Script
A Python script trains a Decision Tree model using the dataset.

2. Data Visualization
Graphs are generated to understand relationships between subject scores.

3. Web Application
A simple Streamlit web interface allows users to enter their scores and receive a recommendation.

The trained model is saved as a .pkl file so it can be reused without retraining each time.

8. Results

The system successfully predicts the subject that a student should focus on based on their scores.

Example:

Input:
Physics: 40
Chemistry: 75
Maths: 50

Output:
Recommended subject: Physics

This demonstrates how machine learning can be used to provide simple educational recommendations.

9. Challenges

Some challenges encountered during the project include:

- Designing a meaningful dataset
- Understanding how classification models work
- Avoiding bias in the dataset
- Integrating the model with a web interface

10. Future Improvements

This project can be improved in several ways:

• Use a larger real-world dataset
• Provide topic-level recommendations instead of only subjects
• Improve model accuracy using advanced algorithms
• Build a mobile or full web application interface

11. Conclusion

This project demonstrates a basic application of machine learning in education.
By analyzing student scores, the system identifies weak subjects and recommends where students should focus their efforts.

Although the model is simple, it illustrates how AI concepts such as data preprocessing, model training,
prediction, and evaluation can be applied to solve real problems.

12. Learning Outcomes

Through this project, the following concepts were learned:

• Machine learning workflow
• Data preprocessing
• Supervised learning and classification
• Model evaluation and prediction
• Building a simple AI application